

SOEN 202 – Web Programming I

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1. Internet, Web and Basic Concepts

Internet

A massive global network of interconnected computers and other devices that communicate using standard protocols (mainly TCP/IP). It allows people and machines all over the world to share information.

World Wide Web (WWW or Web)

A system of interlinked hypertext documents and multimedia resources that are accessed through the Internet using browsers. The Web is one of the most popular services that runs on top of the Internet (others include email, FTP, etc.).

Website

A collection of related web pages and resources (images, videos, scripts, etc.) that are usually under a common domain name and linked together.

Web page

A single document on the Web, normally written in HTML (and often styled with CSS and made interactive with JavaScript). It can contain text, images, links, forms, videos, etc.

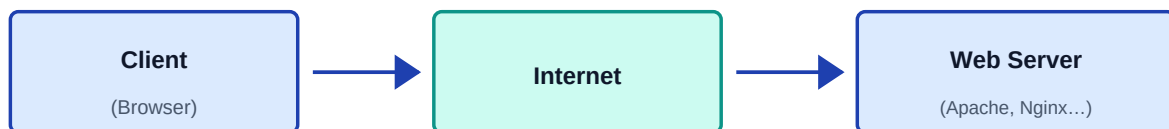
Web browser

Software that requests, downloads and displays web pages. Examples: Chrome, Firefox, Edge, Safari. The browser is the client program that talks to web servers.

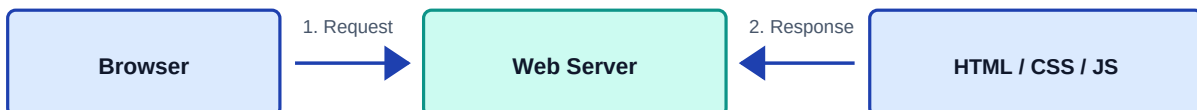
2. How the Web Works (Client–Server)

The Web follows a simple request–response model:

1. You type a URL or click a link in your browser (the client).
2. The browser sends an HTTP request across the Internet to a web server.
3. The server finds (or generates) the requested resource and sends it back as an HTTP response.
4. The browser receives the response and displays the page.



Basic architecture: Client (browser) ↔ Internet ↔ Web Server



Browser sends request → Server processes it → Server sends back the page

Request–Response cycle in simple steps

Web Server

Software (and the computer it runs on) that stores website files and responds to browser requests. Popular examples: Apache, Nginx, Microsoft IIS. The server software listens for requests and returns the correct files or generated content.

3. Uniform Resource Locator (URL)

A URL is the standard address used to locate a specific resource on the Internet. It tells the browser exactly which protocol to use, which server to contact, and which file or path to request.

```
https://www.example.com:443/path/page.html?id=5#section
```

Protocol

Domain / Host

Port

Path

Query + Fragment

Uniform Resource Locator (URL) – the address of a resource on the web

Main parts of a URL:

- **Protocol** – e.g. https:// (secure) or http://
- **Domain / Host** – the human-readable name of the server (www.example.com)
- **Port** – optional number (default 80 for http, 443 for https)
- **Path** – folder and file on the server (/path/page.html)
- **Query string** – extra data sent to the server (?id=5)
- **Fragment** – location inside the page (#section)

4. Static vs Dynamic Websites

Static Website

- Content fixed in HTML files
- Same for every visitor
 - Edited manually
- Fast & simple to host

Dynamic Website

- Content generated on request
- Can change per user / time
- Uses server-side programs
- Databases + scripts (PHP...)

Quick comparison of the two main types of websites

Static website

Pages are fixed files (usually plain HTML). The content is the same for every visitor and only changes when a developer edits and uploads the files. Simple, fast and cheap to host, but limited for interactive or personalised sites.

Dynamic website

Pages are generated on the server when a request arrives. Content can change based on user, time, database data, etc. Built with server-side languages (PHP, Python, Node.js, etc.) and often connected to a database. Examples: online shops, social networks, learning platforms.

5. Website Planning and Design Basics

Before writing code it is important to plan. Key areas include:

- **Purpose** – What is the site for? Who is the audience?
- **Content** – What information and media will be on the site?
- **Structure / Navigation** – How will pages be organised and linked?
- **Layout** – How will content be arranged on each page?
- **Visual design** – Colours, fonts, images, consistency
- **Accessibility** – Making the site usable by people with disabilities

Good page layout follows clear visual hierarchy, enough white space, readable fonts, and consistent navigation so users can find information easily.

6. HTML – HyperText Markup Language

HTML is the standard language used to create the structure and content of web pages. It uses tags (e.g. <h1>, <p>, , <a>) to mark up headings, paragraphs, images, links, tables, forms, etc.

HTML5 introduced semantic elements such as <header>, <nav>, <main>, <article>, <section>, <footer> that describe the meaning of the content, not just its appearance. This helps accessibility and search engines.

Common things you build with HTML:

- Text structure (headings, paragraphs, lists)
- Links and navigation
- Images, audio and video
- Tables
- Forms (input fields, buttons, etc.)

7. CSS – Cascading Style Sheets

CSS is the language used to control the presentation (look and layout) of HTML documents. It separates content (HTML) from design (colours, fonts, spacing, positioning).

Advantages of CSS

- One stylesheet can style many pages → consistent look and easier maintenance
- Smaller HTML files and faster loading
- Better control over layout and responsive design
- Accessibility and device adaptation become easier

Ways to add CSS

- Inline – style attribute on a single element
- Internal – `<style>` block inside the HTML page
- External – separate .css file linked with `<link>` (best practice for real sites)

CSS rules are made of selectors (which elements to style) and declarations (property: value pairs). Example: `h1 { color: navy; font-size: 24px; }`

8. Other Important Topics from the Outline

Search Engines & SEO

Search engines (Google, Bing, etc.) crawl and index web pages so users can find them. Search Engine Optimisation (SEO) is the practice of making pages easier for search engines to understand and rank higher.

Content Management Systems (CMS)

Software that helps people create, manage and publish website content without writing all the code from scratch. Examples: WordPress, Joomla, Drupal. Useful for blogs, company sites and many business websites.

Accessibility

Designing websites so that people with disabilities (visual, hearing, motor, cognitive) can use them. Techniques include proper headings, alt text for images, sufficient colour contrast, keyboard navigation, and semantic HTML.

TCP/IP

The suite of communication protocols that the Internet uses. TCP (Transmission Control Protocol) and IP (Internet Protocol) are the core protocols that allow data to travel reliably between devices on the network.

9. Quick Revision Checklist

- Internet = global network; Web = linked documents and resources on that network.
- Browser (client) requests a page → Web server responds with the page.
- URL is the full address of a resource (protocol + domain + path + ...).
- Static site = fixed files; Dynamic site = generated on request.
- HTML = structure; CSS = presentation; JavaScript = behaviour.
- Always plan purpose, content, structure and accessibility before coding.

These notes expand the class material from SOEN 202 (Silas Momanyi). Use the diagrams to visualise client–server flow, URL parts and static vs dynamic sites when revising for CATs.